PID Controller

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Attitude Control System

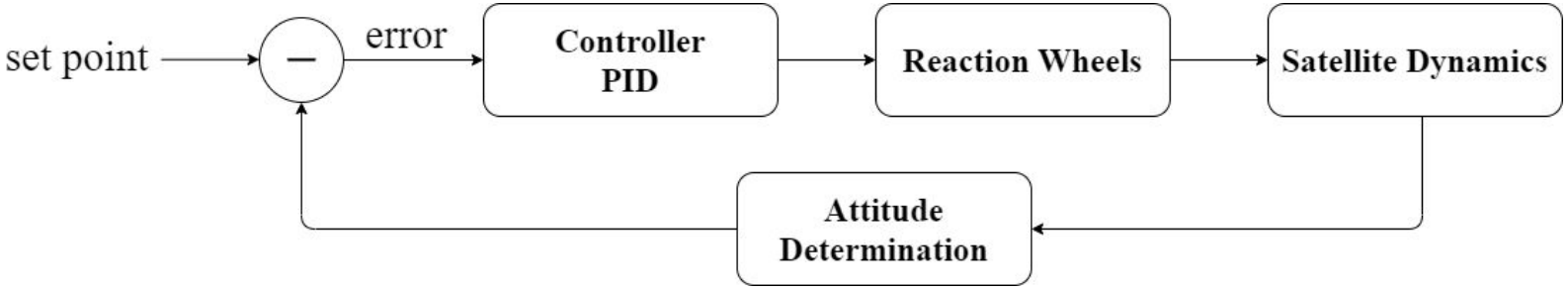

Figure 1. Block diagram of one axis attitude control system

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PID I

$$u(t) = K_p * e(t) + K_i \int e(t)\,dt + K_d \frac{d}{dt} e(t)$$

where,

$u(t) =$ output

$e(t) =$ error = set point - feedback

K_p, K_i, K_d = PID gains

PID II

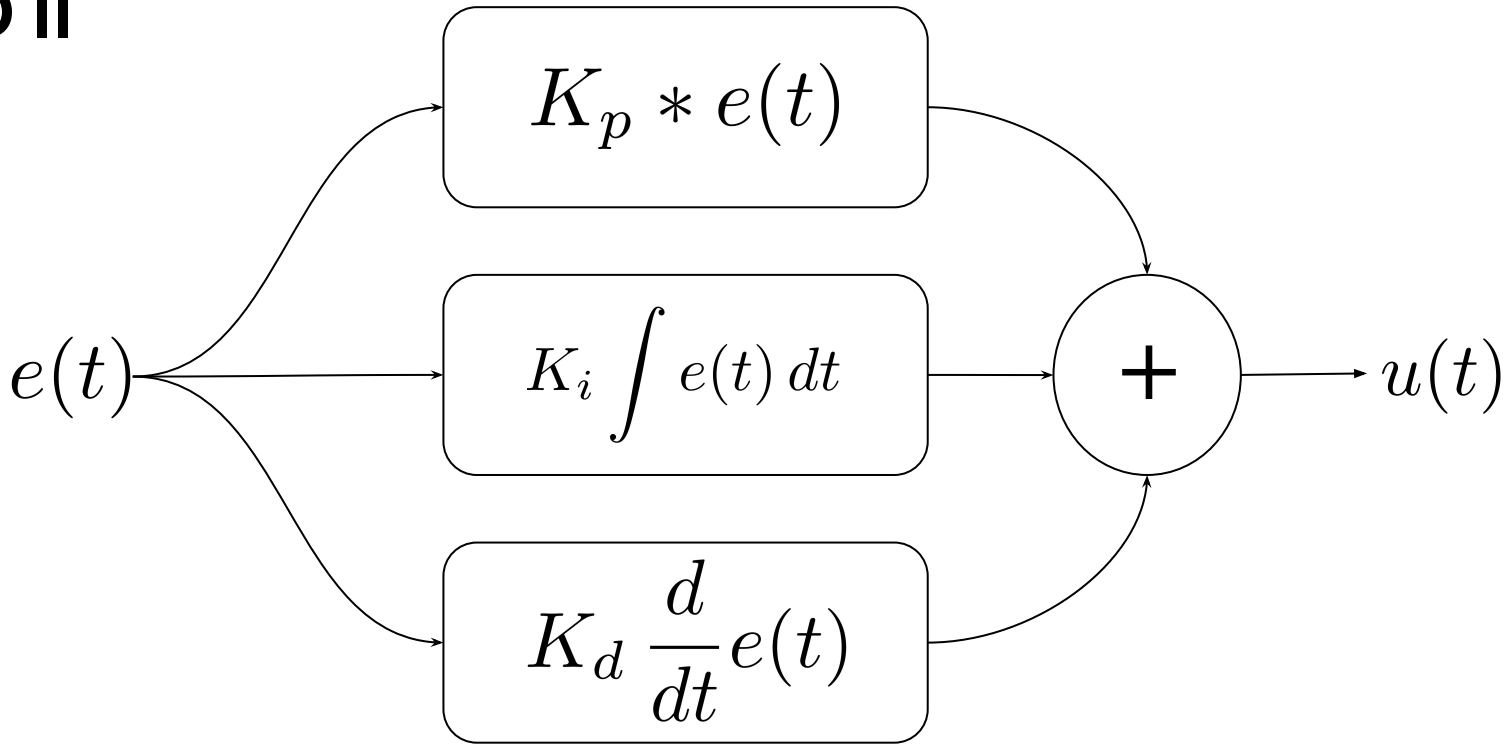

Figure 2. Parallel PID

PID III

$$u_p(t) = K_p * e(t)$$

$$u_i(t) = K_i \int e(t)\, dt$$

$$u_d(t) = K_d \frac{d}{dt} e(t)$$

$$u(t) = u_p(t) + u_i(t) + u_d(t)$$

Q. Digital Implementation ?

A. Discretize

PID in S-domain

$$u_p(s) = K_p\, e(s)$$

$$u_i(s) = \frac{1}{s}\, K_i\, e(s)$$

$$u_d(s) = sK_d\, e(s)$$

$$u(s) = u_p(s) + u_i(s) + u_d(s)$$

Tustin's Approximation

$$s = \frac{2}{T}\frac{z-1}{z+1}$$

Discrete Integration

$$\frac{u_i(s)}{e(s)} = \frac{1}{s} K_i$$

$$\frac{u_i(s)}{e(s)} = K_i \frac{T}{2} \frac{z+1}{z-1}$$

$$\frac{u_i(z)}{e(z)} = K_i \frac{T}{2} \frac{1+z^{-1}}{1-z^{-1}}$$

$$2u_i(z)(1-z^{-1}) = K_i T e(z)(1+z^{-1})$$

$$2(u_i[n] - u_i[n-1]) = K_i T \left(e[n] + e[n-1]\right)$$

$$u_i[n] = \frac{1}{2} K_i T \left(e[n] + e[n-1]\right) + u_i[n-1]$$

Discrete Derivative

$$\frac{u_d(s)}{e(s)} = s\,K_d$$

$$\frac{u_d(s)}{e(s)} = K_d \frac{2}{T} \frac{z-1}{z+1}$$

$$\frac{u_i(z)}{e(z)} = K_d \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}$$

$$Tu_d(z)(1+z^{-1}) = 2K_d e(z)(1-z^{-1})$$

$$T(u_d[n] + u_d[n-1]) = 2K_d\,(e[n] - e[n-1])$$

$$u_d[n] = \frac{2}{T} K_d\,(e[n] - e[n-1]) - u_d[n-1]$$

Discrete PID

$$
\begin{aligned}
u[n] &= u_p[n] + u_i[n] + u_d[n] \\
&= K_p * e[n] + \\
&\quad \frac{1}{2} K_i T \left(e[n] + e[n-1] \right) + u_i[n-1] + \\
&\quad \frac{2}{T} K_d \left(e[n] - e[n-1] \right) - u_i[n-1]
\end{aligned}
$$

Controller Performance Metrics

1. Maximum overshoot (c_p)
2. Time to maximum overshoot (t_p)
3. Time to zero error (t_o)
4. Rise time (t_r)
5. Settling time (t_s)
6. Frequency of oscillation of the transient

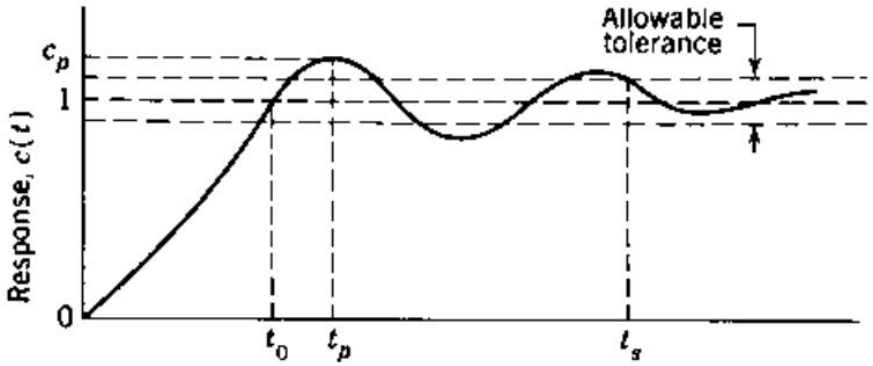

Figure 3. Control system performance metrics

PID Tuning

Effects of *increasing* a parameter independently

Parameter	Rise time	Overshoot	Settling time	Steady-state error	Stability
K_p	Decrease	Increase	Small change	Decrease	Degrade
K_i	Decrease	Increase	Increase	Eliminate	Degrade
K_d	Minor change	Decrease	Decrease	No effect in theory	Improve if K_d small

Q. Exactly What Does Controller Do?

A. Changes the dynamics itself

Q. When to use PID controller?

A. For linear system

References

1. Online PID simulator: https://sites.google.com/site/fpgaandco/pid
2. Brian Douglas: https://www.youtube.com/channel/UCq0imsn84ShAe9PBOFnoIrg
3. Phil's lab: https://www.youtube.com/watch?v=zOByx3Izf5U&t=425s
4. Astrom, Hagglund - Advanced PID Control (2006)

Thank you !!